



Scottsdale EZ Predictive Analysis

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Capstone Team 2



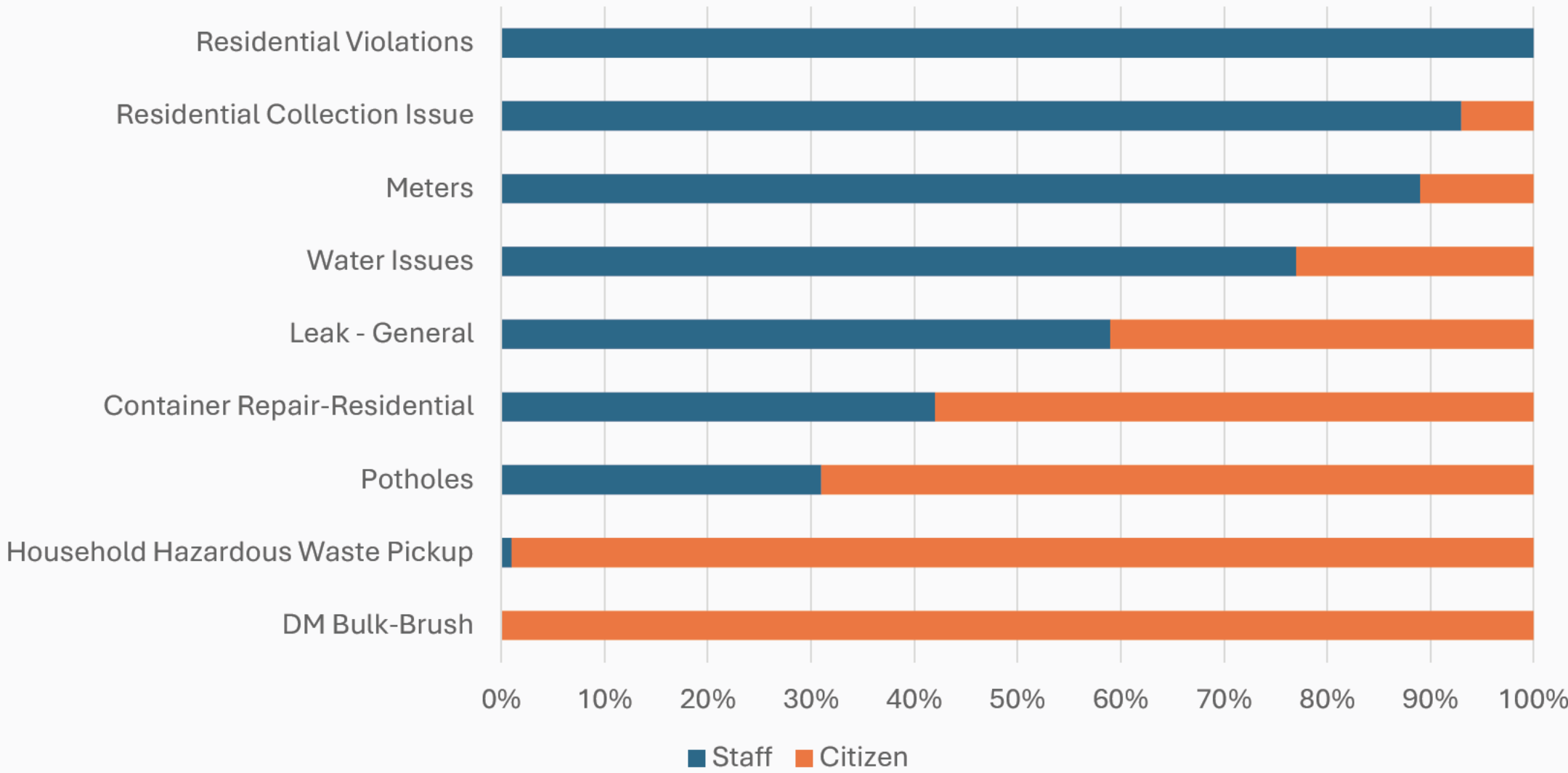
Problem Statement

This project adds a predictive layer to ScottsdaleEZ, the citizen request system, aiding with human-in-the-loop decisions

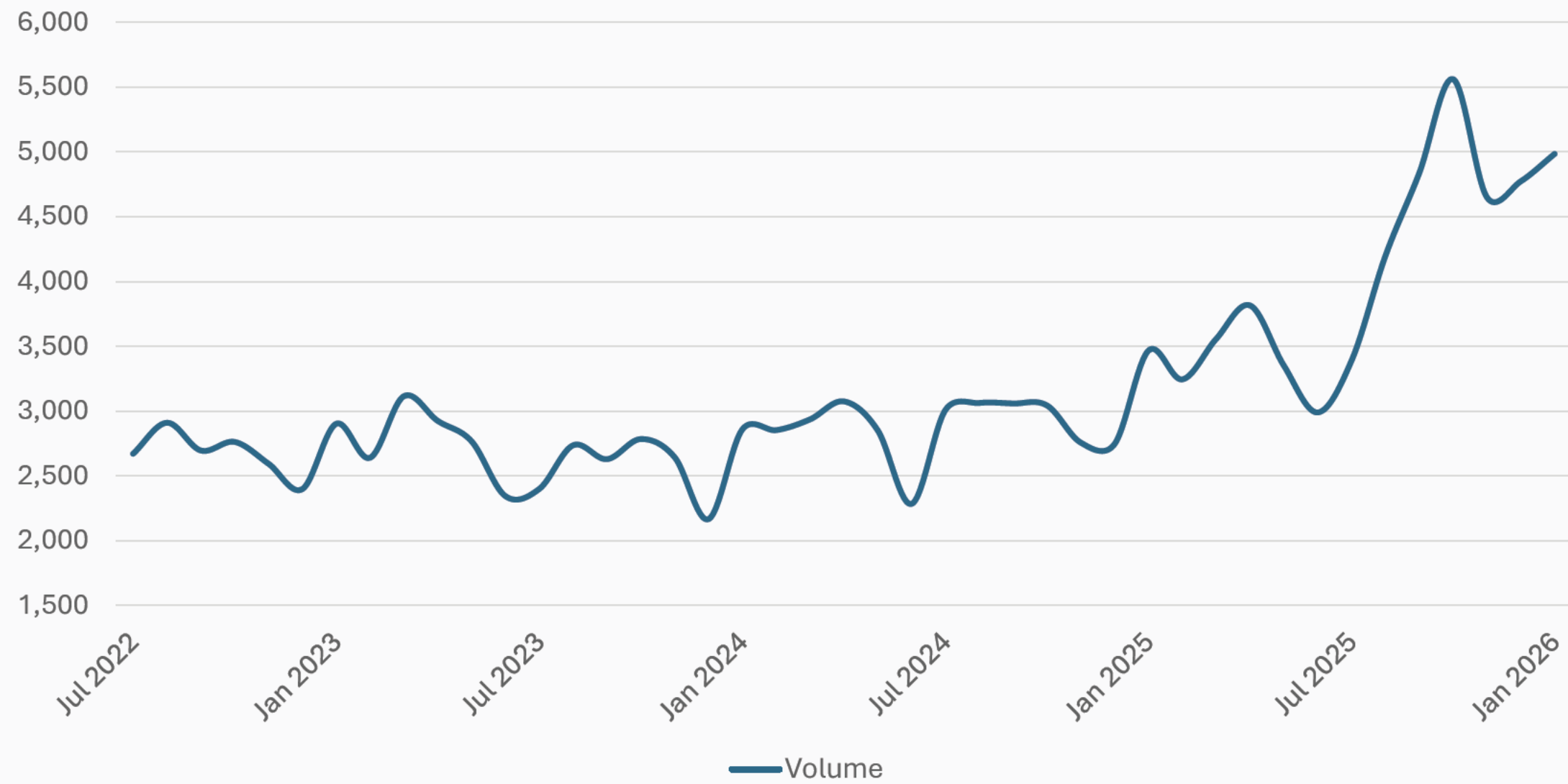
Data Foundation

≈ 200K Total Cases (Jul '22 - Jan '26) 52% Staff 48% Citizens 256 Request types 31 Workgroups ≈ 139K Cases Used for Modeling after exclusions

Staff vs. Citizen Submission Mix by Request Type



Monthly Service Request Volume (Jul 2022 – Jan 2026)



Cases Post Launch
206K

Cases per Month
3-5K

Model Used
LightGBM

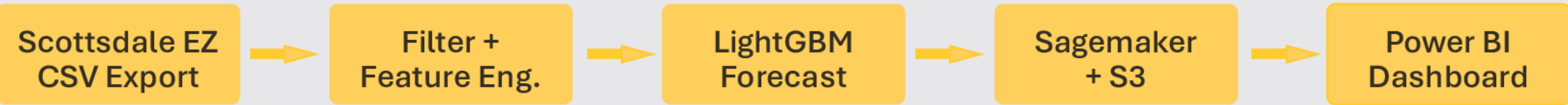
Features Used
Large volumes, rolling summaries, seasonal variables

Project Output
Data analysis, Power BI Dashboard

Aggregate Error
45%*

Methods

- Forecasting unit: ZIP code × request type × month
- Target: next-month service request volume
- Features included lagged volumes, rolling averages, and seasonal variables
- Time-based train/test split: trained on July 2022 – June 2025, tested on July 2025 onward — no data leakage across the temporal boundary

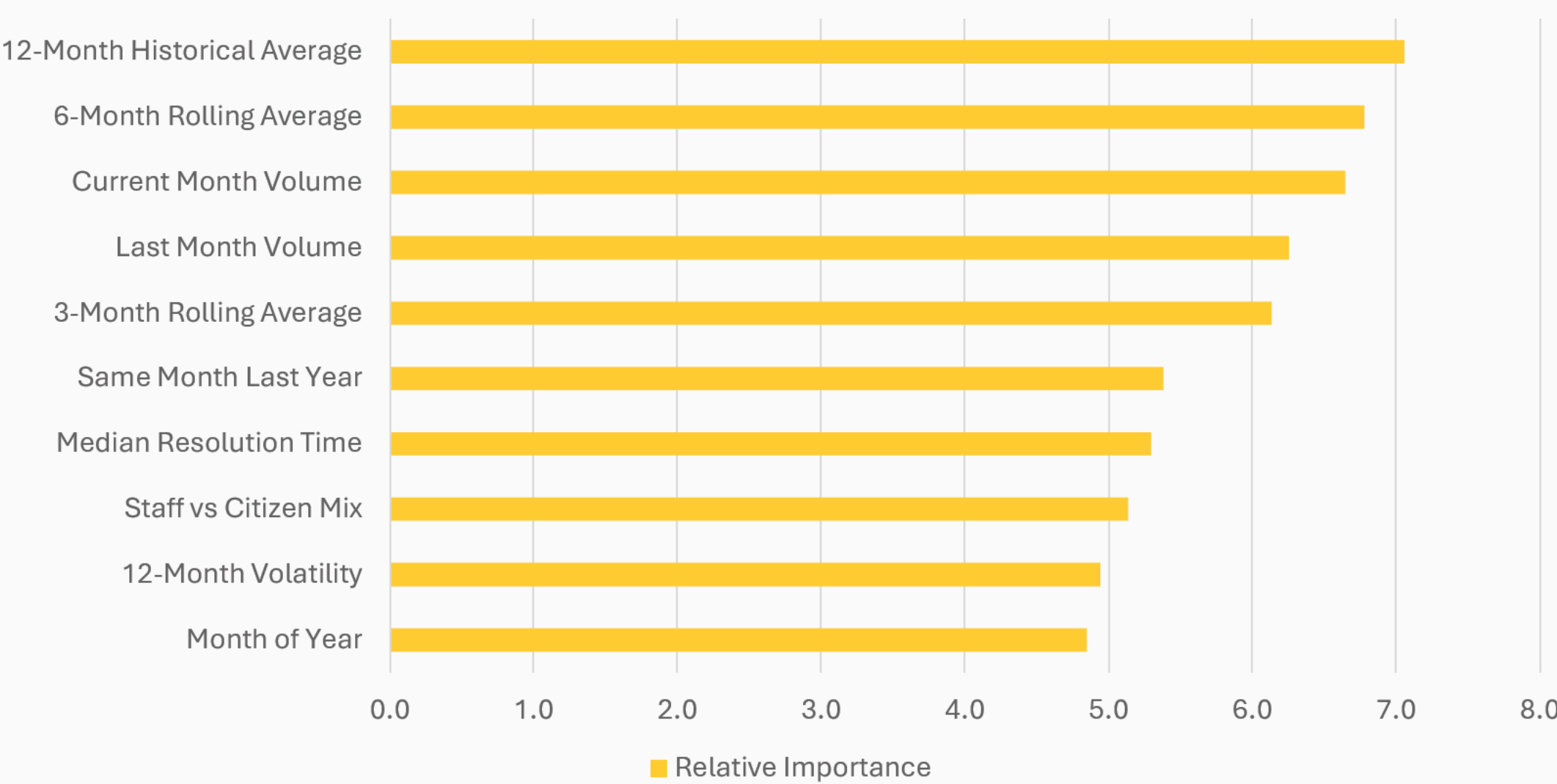


Multiple forecasting models were tested against held-out future data.

LightGBM was selected because it:

- Delivered the strongest predictive accuracy on unseen data
- Handled sparse, mixed, and nonlinear municipal data effectively
- Generalized consistently to future months without overfitting
- Offered the best balance of accuracy, stability, and deployability

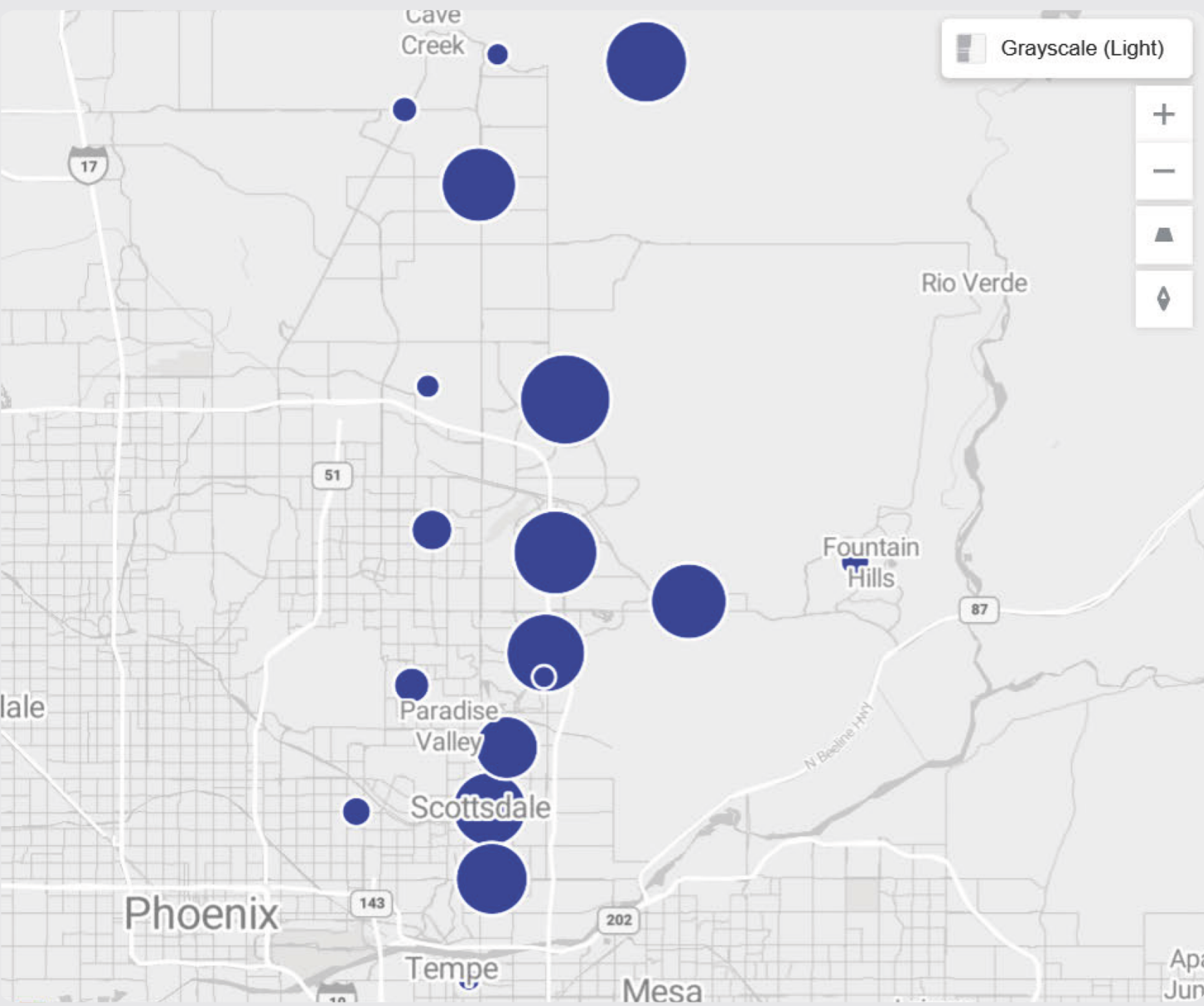
Feature Importance (Gain-Based, Log Scale)



* City demand forecasting is an emerging research area. Xu et al. (2017), among the first peer-reviewed studies on the problem, found that data sparsity fundamentally limits forecast precision across models. Our ~45% aggregate error combined with human-on-the-loop decision making will help early warning for city operations

Power BI Dashboard

- The ScottsdaleEZ Power BI dashboard translates model outputs into an operational decision interface
- Department heads can view predicted hotspots by ZIP code, ranked issue lists, SLA performance, and year-over-year comparisons, enabling proactive staffing and prioritization before issues arise.



Power BI Hotspot Dashboard Shows where next-month demand is expected

Recommendations

- Proactive review of the ScottsdaleEZ data as a way to predict Citizen needs
- Promote citizen adoption of ScottsdaleEZ to increase data volume and signal quality
- Differentiate ZIP code and Staff use in data input to improve data quality, possible addition of weather and other external driver data
- Monitor model performance post-deployment and retrain as new data accumulates